pingall:

Screenshot:

```
mininet> pingall
*** Ping: testing ping reachability
h101 -> h102 h103 h104 X X X X h_server h_trust X
h102 -> h101 h103 h104 X X X X h_server h_trust X
h103 -> h101 h102 h104 X X X X h_server h_trust X
h104 -> h101 h102 h103 X X X X h_server h_trust X
h201 -> X X X X h202 h203 h204 h_server X X
h202 -> X X X X h201 h203 h204 h_server X X
h203 -> X X X X h201 h202 h204 h_server X X
h204 -> X X X X h201 h202 h203 h_server X X
h_server -> h101 h102 h103 h104 h201 h202 h203 h204 X X
h_trust -> h101 h102 h103 h104 X X X X X h_untrust
h_untrust -> X X X X X X X X X h_trust
*** Results: 54% dropped (50/110 received)
mininet>
```

Explanation:

The “pingall” command shows the hosts and their reachability from the other hosts. As per the instructions, all hosts are able to communicate except with a few constraints for specific hosts.

The constraint for the untrusted host is that it cannot send ICMP and IP traffic to Host 101-104, 201-204, or the LLM Server, implying that it can only send said traffic to the trusted host. As seen in the screenshot, it drops all packets (shown through the Xs) except for the trusted host.

The constraint for the trusted host, is that it cannot send IP and ICMP traffic to Host 201-204, or the LLM server, implying it can only send to the untrusted host and Host 101-104. As seen in the screenshot, it drops all packets (shown through the Xs) except for the untrusted host and Host 101-104.

The first 4 hosts in the screenshot, Hosts 101-104, show that it does not communicate with Hosts 201-204 and the Untrusted Host as shown through the Xs. The second set of 4 hosts in the screenshot, Hosts 201-204 show that it does not communicate with Hosts 101-104 and the Untrusted Host as shown through the Xs.

dump:

Screenshot:

```
mininet> dump
<Host h101: h101-eth0:128.114.1.101 pid=1860>
<Host h102: h102-eth0:128.114.1.102 pid=1864>
<Host h103: h103-eth0:128.114.1.103 pid=1866>
<Host h104: h104-eth0:128.114.1.104 pid=1868>
<Host h201: h201-eth0:128.114.2.201 pid=1870>
<Host h202: h202-eth0:128.114.2.202 pid=1872>
<Host h203: h203-eth0:128.114.2.203 pid=1874>
<Host h204: h204-eth0:128.114.2.204 pid=1876>
<Host h_server: h_server-eth0:128.114.3.178 pid=1878>
<Host h_trust: h_trust-eth0:192.47.38.109 pid=1880>
<Host h_untrust: h_untrust-eth0:108.35.24.113 pid=1882>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None,s1-eth3:None pid=1887>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=1890>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=1893>
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None,s4-eth4:None,
s4-eth5:None,s4-eth6:None,s4-eth7:None pid=1896>
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=1899>
<OVSSwitch s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None pid=1902>
<RemoteController c0: 127.0.0.1:6633 pid=1854>
mininet>
```

Explanation:

The “dump” command is executed as seen in the screenshot above. As seen, all the hosts, the LLM server, the trusted host, and untrusted hosts are all present. The switches are shown as well. Additionally there are six switches, representing floor 1 switch 1 (s1), floor 1 switch 2 (s2), floor 2 switch 1 (s3), floor 2 switch 2 (s5), the core switch (s4), and the data center switch (s6).

links:

Screenshot:

```
mininet> links
s1-eth1<->h101-eth0 (OK OK)
s1-eth2<->h102-eth0 (OK OK)
s1-eth3<->s4-eth1 (OK OK)
s2-eth1<->h103-eth0 (OK OK)
s2-eth2<->h104-eth0 (OK OK)
s2-eth3<->s4-eth2 (OK OK)
s3-eth1<->h201-eth0 (OK OK)
s3-eth2<->h202-eth0 (OK OK)
s3-eth3<->s4-eth3 (OK OK)
s4-eth6<->h_trust-eth0 (OK OK)
s4-eth7<->h_untrust-eth0 (OK OK)
s5-eth1<->h203-eth0 (OK OK)
s5-eth2<->h204-eth0 (OK OK)
s5-eth3<->s4-eth4 (OK OK)
s6-eth1<->h_server-eth0 (OK OK)
s6-eth2<->s4-eth5 (OK OK)
mininet>
```

Explanation:

There are six switches, representing floor 1 switch 1 (s1), floor 1 switch 2 (s2), floor 2 switch 1 (s3), floor 2 switch 2 (s5), the core switch (s4), and the data center switch (s6). This command shows that Host 101 and Host 102 are connected to s1. It shows that Host 103 and Host 104 are connected to s2. It also shows that Host 201 and Host 202 are connected to s3. It shows that Host 203 and Host 204 are connected to s5. Trusted host and Untrusted host are shown to be connected to s4, the core switch, directly. The switch s1, s2, s3, s5, and s6 is shown to be connected to s4, the core switch. The LLM server is shown to be connected to s6.

nodes:

Screenshot:

```
mininet> nodes
available nodes are:
c0 h101 h102 h103 h104 h201 h202 h203 h204 h_server h_trust h_untrust s1 s2 s3 s
4 s5 s6
mininet>
```

Explanation:

The “nodes” command shows the creation of devices. This satisfies the check to see if the devices are successfully created. As seen in the screenshot, Hosts101-104, Hosts 201-204, LLM server, Trusted Host, and Untrusted Host are successfully created, as well as the switches.

net:

Screenshot:

```
mininet> net
h101 h101-eth0:s1-eth1
h102 h102-eth0:s1-eth2
h103 h103-eth0:s2-eth1
h104 h104-eth0:s2-eth2
h201 h201-eth0:s3-eth1
h202 h202-eth0:s3-eth2
h203 h203-eth0:s5-eth1
h204 h204-eth0:s5-eth2
h_server h_server-eth0:s6-eth1
h_trust h_trust-eth0:s4-eth6
h_untrust h_untrust-eth0:s4-eth7
s1 lo:  s1-eth1:h101-eth0 s1-eth2:h102-eth0 s1-eth3:s4-eth1
s2 lo:  s2-eth1:h103-eth0 s2-eth2:h104-eth0 s2-eth3:s4-eth2
s3 lo:  s3-eth1:h201-eth0 s3-eth2:h202-eth0 s3-eth3:s4-eth3
s4 lo:  s4-eth1:s1-eth3 s4-eth2:s2-eth3 s4-eth3:s3-eth3 s4-eth4:s5-eth3 s4-eth5:
s6-eth2 s4-eth6:h_trust-eth0 s4-eth7:h_untrust-eth0
s5 lo:  s5-eth1:h203-eth0 s5-eth2:h204-eth0 s5-eth3:s4-eth4
s6 lo:  s6-eth1:h_server-eth0 s6-eth2:s4-eth5
c0
mininet>
```

Explanation:

The “net” command shows the hosts 101-104 and the hosts 201-204, along with the switches they are connected to. There are six switches, representing floor 1 switch 1 (s1), floor 1 switch 2 (s2), floor 2 switch 1 (s3), floor 2 switch 2 (s5), the core switch (s4), and the data center switch (s6). This command shows that Host 101 and Host 102 are connected to s1. It shows that Host 103 and Host 104 are connected to s2. It also shows that Host 201 and Host 202 are connected to s3. It shows that Host 203 and Host 204 are connected to s5. Trusted host and Untrusted host are shown to be connected to s4, the core switch, directly. The LLM server is shown to be connected to s6. The switch s1, s2, s3, s5, and s6 is shown to be connected to s4, the core switch.

untrusted host check:

Screenshot:

```
mininet> h_untrust ping -c 3 101
connect: Invalid argument
mininet> h_untrust ping -c 3 h101
PING 128.114.1.101 (128.114.1.101) 56(84) bytes of data.

--- 128.114.1.101 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet> h_untrust ping -c 3 h102
PING 128.114.1.102 (128.114.1.102) 56(84) bytes of data.

--- 128.114.1.102 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet> h_untrust ping -c 3 h103
PING 128.114.1.103 (128.114.1.103) 56(84) bytes of data.

--- 128.114.1.103 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2008ms

mininet> h_untrust ping -c 3 h104
PING 128.114.1.104 (128.114.1.104) 56(84) bytes of data.

--- 128.114.1.104 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet>
```

```
mininet> h_untrust ping -c 3 h201
PING 128.114.2.201 (128.114.2.201) 56(84) bytes of data.

--- 128.114.2.201 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2015ms

mininet> h_untrust ping -c 3 h202
PING 128.114.2.202 (128.114.2.202) 56(84) bytes of data.

--- 128.114.2.202 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2015ms

mininet> h_untrust ping -c 3 h203
PING 128.114.2.203 (128.114.2.203) 56(84) bytes of data.

--- 128.114.2.203 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2015ms

mininet> h_untrust ping -c 3 h204
PING 128.114.2.204 (128.114.2.204) 56(84) bytes of data.

--- 128.114.2.204 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2016ms

mininet>
```

```
mininet> h_untrust ping -c 3 h_trust
PING 192.47.38.109 (192.47.38.109) 56(84) bytes of data.
64 bytes from 192.47.38.109: icmp_seq=1 ttl=64 time=42.4 ms
64 bytes from 192.47.38.109: icmp_seq=2 ttl=64 time=0.880 ms
64 bytes from 192.47.38.109: icmp_seq=3 ttl=64 time=0.070 ms

--- 192.47.38.109 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.070/14.474/42.473/19.801 ms
mininet>
```

Explanation:

As shown in the first screenshot above, the untrusted host cannot send ICMP traffic to Host 101-104. This is shown through the “100% packet loss”. As shown in the second screenshot above, the untrusted host cannot send ICMP traffic to Host 201-204. This is shown through the “100% packet loss”. It can reach the trusted host, as shown by the “0% packet loss”.

trusted host check:

Screenshot:

```
mininet> h_trust ping -c 3 h101
PING 128.114.1.101 (128.114.1.101) 56(84) bytes of data.
64 bytes from 128.114.1.101: icmp_seq=1 ttl=64 time=40.0 ms
64 bytes from 128.114.1.101: icmp_seq=2 ttl=64 time=0.916 ms
64 bytes from 128.114.1.101: icmp_seq=3 ttl=64 time=0.058 ms

--- 128.114.1.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.058/13.663/40.015/18.636 ms
mininet> h_trust ping -c 3 h102
PING 128.114.1.102 (128.114.1.102) 56(84) bytes of data.
64 bytes from 128.114.1.102: icmp_seq=1 ttl=64 time=44.8 ms
64 bytes from 128.114.1.102: icmp_seq=2 ttl=64 time=0.719 ms
64 bytes from 128.114.1.102: icmp_seq=3 ttl=64 time=0.056 ms

--- 128.114.1.102 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.056/15.214/44.868/20.970 ms
mininet> h_trust ping -c 3 h103
PING 128.114.1.103 (128.114.1.103) 56(84) bytes of data.
64 bytes from 128.114.1.103: icmp_seq=1 ttl=64 time=54.5 ms
64 bytes from 128.114.1.103: icmp_seq=2 ttl=64 time=1.03 ms
64 bytes from 128.114.1.103: icmp_seq=3 ttl=64 time=0.056 ms

--- 128.114.1.103 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 0.056/18.555/54.577/25.474 ms
mininet> h_trust ping -c 3 h104
PING 128.114.1.104 (128.114.1.104) 56(84) bytes of data.
64 bytes from 128.114.1.104: icmp_seq=1 ttl=64 time=48.5 ms
64 bytes from 128.114.1.104: icmp_seq=2 ttl=64 time=0.672 ms
64 bytes from 128.114.1.104: icmp_seq=3 ttl=64 time=0.053 ms

--- 128.114.1.104 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.053/16.434/48.577/22.729 ms
mininet>
```

```
mininet> h_trust ping -c 3 h201
PING 128.114.2.201 (128.114.2.201) 56(84) bytes of data.

--- 128.114.2.201 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet> h_trust ping -c 3 h202
PING 128.114.2.202 (128.114.2.202) 56(84) bytes of data.

--- 128.114.2.202 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2007ms

mininet> h_trust ping -c 3 h203
PING 128.114.2.203 (128.114.2.203) 56(84) bytes of data.

--- 128.114.2.203 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2015ms

mininet> h_trust ping -c 3 h204
PING 128.114.2.204 (128.114.2.204) 56(84) bytes of data.

--- 128.114.2.204 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2018ms

mininet>
```

```
mininet> h_trust ping -c 3 h_untrust
PING 108.35.24.113 (108.35.24.113) 56(84) bytes of data.
64 bytes from 108.35.24.113: icmp_seq=1 ttl=64 time=27.5 ms
64 bytes from 108.35.24.113: icmp_seq=2 ttl=64 time=0.589 ms
64 bytes from 108.35.24.113: icmp_seq=3 ttl=64 time=0.055 ms

--- 108.35.24.113 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 0.055/9.402/27.562/12.842 ms
mininet> h_trust ping -c 3 h_server
PING 128.114.3.178 (128.114.3.178) 56(84) bytes of data.

--- 128.114.3.178 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet>
```

Explanation:

The trusted host, as shown in the screenshot on the top and bottom, can ping Host 101-104 and Untrusted Host. This is supported through the “0% packet loss”. As seen in the screenshot on the middle and bottom, the trusted host cannot send ICMP traffic to Host 201-204 or LLM Server, as supported by the “100% packet loss” message.

LLM server check:

Screenshot:

```
mininet> h_trust ping -c 3 h_server
PING 128.114.3.178 (128.114.3.178) 56(84) bytes of data.

--- 128.114.3.178 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 1999ms

mininet> h_untrust ping -c 3 h_server
PING 128.114.3.178 (128.114.3.178) 56(84) bytes of data.

--- 128.114.3.178 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2016ms
```

Explanation:

The screenshot above shows that the untrusted host and the trusted host cannot send traffic to the LLM Server. This is supported by the “100% packet loss” and “0 received” messages referring to the packets that were not received.

host 101-104 to host 201-204 check:

Screenshot:

```
mininet> h101 ping -c 3 h201
PING 128.114.2.201 (128.114.2.201) 56(84) bytes of data.

--- 128.114.2.201 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2017ms

mininet> h204 ping -c 3 h104
PING 128.114.1.104 (128.114.1.104) 56(84) bytes of data.

--- 128.114.1.104 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2002ms

mininet>
```

Explanation:

The above screenshot shows examples that prove that Hosts 101-104 cannot send traffic to Hosts 201-204. This is shown through the examples that result in 100% packet loss.